Schlumberger

Company: BOB PT. BUMI SIAK PUSAKO - PERTAMINA HULU

Well: PDD-M01 (# 156)

Field: PEDADA

Rig Name: APS-552 Country: INDONESIA

Rig Name: APS-552
Field: PEDADA
Location: PEDADA
Well: PDD-M01 (# 156)
Company: BOB PT. BUMI SIAK PUSAKO - PERTAMINA H

CBL-VDL-GR-CCL LOG

17-OCT-2014

SCALE 1:200

Location:			
PEDADA		Elev.: K.B.	32.71 ft
X: 843,352.81 mE		G.L.	16.71 ft
Y: 124,383.58 mN		D.F.	31.71 ft
Permanent Datum:	Ground Level	Elev.:	16.71 ft
Log Measured From:	Kelly Bushing	16.00 ft	above Perm.Datum
Drilling Measured From:	Kelly Bushing		

Province:	Max.Hole Deviation	Longitude:	Latitude:
RIAU	2 deg	102° 5' 4.93" E	1° 7' 25.29" N

Logging Date		17-Oct-2014	
Run Number		1	
Depth Driller		676.00 ft	
Schlumberger Depth		672.00 ft	
Bottom Log Interval		666.75 ft	
Top Log Interval		275.00 ft	
Casing Fluid Type		KCL Polymer	
Salinity			
Density		8.8 lbm/gal	
Fluid Level		100.00 ft	
BIT/CASING/TUBING STRING			
Bit Size		8.50 in	
From		275.00 ft	
To		672.00 ft	
Casing/Tubing Size		7 in	
Weight		23 lbm/ft	
Grade		K55	
From		0.00 ft	
To		672.00 ft	
Max Recorded Temperatures		100 degF	
Logger on Bottom	Time	17-Oct-2014 19:30:35	
Unit Number	Location:	3151 ASA/ING/CTX	
Recorded By		Pandapotan	
Witnessed By		Mr. Rahmat Purba, Mr. Roby Oksuanandi	

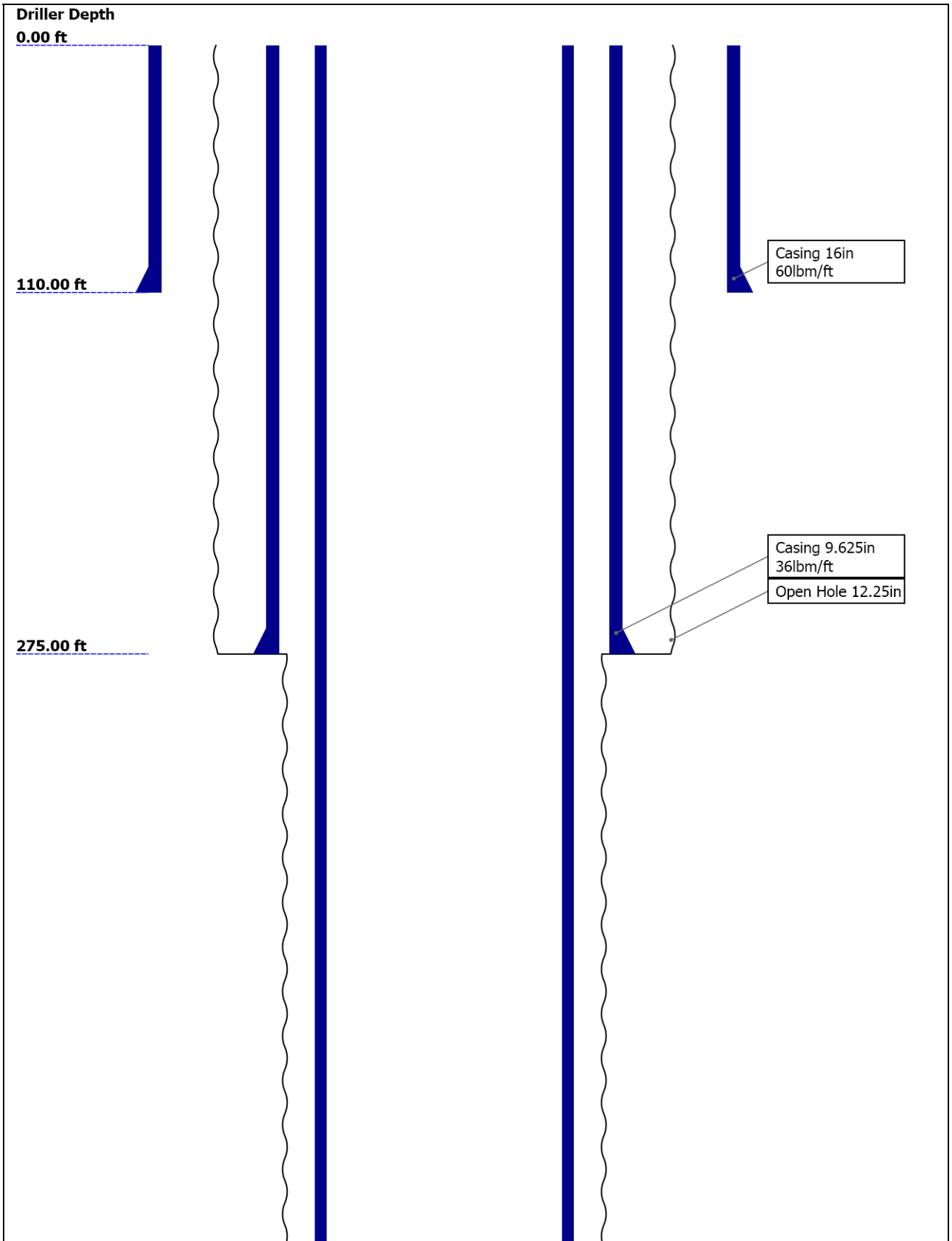
Disclaimer

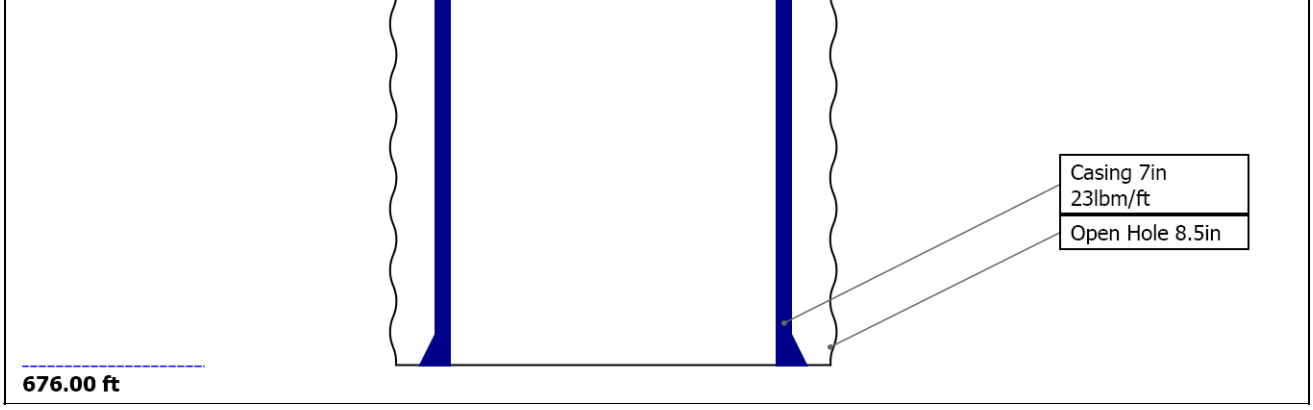
THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

Contents

1. Header
2. Disclaimer
3. Contents
4. Well Sketch
5. Borehole Size/Casing/Tubing Record
6. Operational Run Summary
7. Remarks and Equipment Summary
8. Depth Summary
9. Survey Record
10. Composite 3 Main Log 1:200
    10.1 Integration Summary
    10.2 Composite Summary
    10.3 Log (Sonic CBL with VDL)
11. 1 Repeat Analysis 1:200
    11.1 Composite Summary
    11.2 Log (Sonic CBL with VDL RA)
12. Tail

Well Sketch


Driller Depth
0.00 ft
110.00 ft
275.00 ft
Casing 16in
60lbm/ft
Casing 9.625in
36lbm/ft
Open Hole 12.25in



Borehole Size/Casing/Tubing Record

Bit						
Bit Size (in)	12.25	8.5				
Top Driller (ft)	0	275				
Top Logger (ft)	0	275				
Bottom Driller (ft)	275	676				
Bottom Logger (ft)	275	672				
Casing						
Size (in)	16	9.625	7			
Weight (lbm/ft)	60	36	23			
Inner Diameter (in)	15.298	8.921	6.366			
Grade	K55	K55	K55			
Top Driller (ft)	0	0	0			
Top Logger (ft)	0	0	0			
Bottom Driller (ft)	110	275	676			
Bottom Logger (ft)	110	275	672			

Operational Run Summary

Parameter (unit)	1					
Date Log Started	17-Oct-2014					
Time Log Started	18:59:05					
Date Log Finished	17-Oct-2014					
Time Log Finished	19:36:43					
Top Log Interval (ft)	275.00					
Bottom Log Interval (ft)	666.75					
Total Depth (ft)	669.00					
Max Hole Deviation (deg)	2.00					
Azimuth of Max Deviation (deg)	35.75					
Bit Size (in)	8.500					
Logging Unit Number	3151					
Logging Unit Location	ASA/ING/CTX					
Recorded By	Pandapotan					
Witnessed By	Mr. Rahmat Purba, Mr. Roby					

	Oksuanandi					
Service Order Number	CPP-1410007					

Remarks and Equipment Summary

1: Toolstring

Equip name	Length	MP name	Offset
LEH-QT LEH-QT	34.01		
CAL-YA CAL-YA	31.1		
		CCL	30.31
EDTC-B EDTH-B EDTG-A EDTC-B	27.6		
		CTEM	24.1
		ACCZ	0.00
		HV	0.00
		Gamma Ray	22.23
		TelStatus	21.1
DSLT-H:8145 ECH-KH DSLC-H:8145 SLS-E:1025	21.1		
		CBL 3ft	8.62
		Upper-Near	8.62
		VDL 5ft	7.62
		Upper-Far	7.62
		Delta-T	6.25
		Lower-Far	4.88
		Lower-Near	3.88
		SLS-E	0.46
BNS-STD	0.46	**Head Tension** TOOL_ZERO	

Lengths are in ft
Maximum Outer Diameter = 6.250 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

1: Remarks

1. Log objective: Cement Bond Evaluation
2. Tool string run as per tool sketch
3. This is subsequent trip in hole. Log was correlated to PEX-HALS MD LOG dated 15-Oct-2014 recorded by Schlumberger.
4. TDL tagged at 672 FT. Free pipe not founded.
5. Any missing header data due to unavailable at wellsite

Depth Summary

	1		
Depth Measuring Device			
Type	IDW-B		
Serial Number	1925		

Calibration Date	10-May-2014		
Calibrator Serial Number	1		
Calibration Cable Type	7-46 ZVXS		
Wheel Correction 1	-5		
Wheel Correction 2	-5		

Tension Device

Type	CMTD-B/A		
Serial Number	2386		
Calibration Date	10-Oct-2014		
Calibrator Serial Number	1106		
Number of Calibration Points	10		
Calibration Root Mean Square Error	30		
Calibration Peak Error	41		

Logging Cable

Type	7-46ZV-XS		
Serial Number	77169		
Length	8200.00 ft		
Conveyance Type	Wireline		
Rig Type	DRILLING		

1:Depth Control Parameters

Log Sequence	Subsequent Trip To the Well
Reference Log Name	PEX-HALS MD LOG
Reference Log Run Number	1
Reference Log Date	15-Oct-2014
Subsequent Trip Down Log Correction	0.10 ft

Depth Control Remarks

Schlumberger Wireline Depth Measurement and Control Standard (7-Apr-2010) and Procedures (21-Dec-2010) followed

IDW used as primary depth control measurement device; Z-chart used as secondary depth control reference.

This is subsequent trip in hole. Log was correlated to PEX-HALS MD LOG dated 15-Oct-2014 recorded by Schlumberger.

Survey Record

Survey Calculation

Method :	Minimum Radius of Curvature	DLS Method :	Lubinski
North Reference :	True North	Total Correction Formula :	Magnetic Dec

Rig Location

Latitude :	1° 7' 25.29" N	Longitude :	102° 5' 4.93" E

Tie In Point

Measured Depth:	0.00 ft	Inclination:	0.00 deg	Azimuth:	0.00 deg
True Vertical Depth:	0.00 ft	North Displacement:	0.00 ft	East Displacement:	0.00 ft

Survey Quality Index

9 : Manual 28 : Tie-In Point

Survey Correction Index

0 : No correction

Survey Description Index

0 : Not Flagged Survey

Seq	MD (ft)	Incl (deg)	Azim (deg)	Course (ft)	TVD (ft)	V Sec (ft)	N/ -S (ft)	E/ -W (ft)	Closure (ft)	at Azim (deg)	DLS deg/100ft	Tool Type	QI	CI	DI
1	0.00	0.00	0.00	- - - -	0.00	0.00	0.00	0.00	0.00	90.00	0.00	TIP	28	0	0
2	347.00	1.46	39.61	347.00	346.96	3.41	3.41	2.82	4.43	39.61	0.42	Other	9	0	0
3	378.00	2.00	35.75	31.00	377.95	4.15	4.15	3.39	5.35	39.22	1.78	Other	9	0	0
4	408.00	1.98	27.46	30.00	407.93	5.03	5.03	3.93	6.40	37.99	0.96	Other	9	0	0
5	438.00	1.44	17.97	30.00	437.92	5.85	5.85	4.29	7.25	36.22	2.03	Other	9	0	0
6	468.00	0.99	1.45	30.00	467.91	6.47	6.47	4.41	7.84	34.28	1.89	Other	9	0	0
7	498.00	0.70	302.59	30.00	497.91	6.83	6.83	4.26	8.04	31.97	2.89	Other	9	0	0
8	529.00	1.17	300.74	31.00	528.90	7.09	7.09	3.83	8.07	28.37	1.52	Other	9	0	0
9	560.00	1.27	311.31	31.00	559.90	7.48	7.48	3.30	8.17	23.80	0.79	Other	9	0	0
10	587.00	0.85	333.97	27.00	586.89	7.86	7.86	2.99	8.40	20.82	2.17	Other	9	0	0
11	620.00	1.49	34.03	33.00	619.89	8.43	8.43	3.12	8.99	20.30	3.93	Other	9	0	0
12	652.00	1.94	69.51	32.00	651.87	8.97	8.97	3.86	9.78	23.29	3.53	Other	9	0	0

13	715.00	2.83	139.36	63.00	714.83	8.16	8.16	5.87	10.04	35.74	4.49	Other	9	0	0

Composite 3

Main Log 1:200

Composite Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	DSC Mode	Depth Shift	Include Parallel Data
1	Log[1]:Down	Down	30.53 ft	338.62 ft	17-Oct-2014 7:04:11 PM	17-Oct-2014 7:12:57 PM	ON	-0.60 ft	No
1	Log[3]:Up	Up	232.97 ft	674.82 ft	17-Oct-2014 7:19:23 PM	17-Oct-2014 7:28:15 PM	ON	-0.60 ft	No

All depths are referenced to toolstring zero

Log

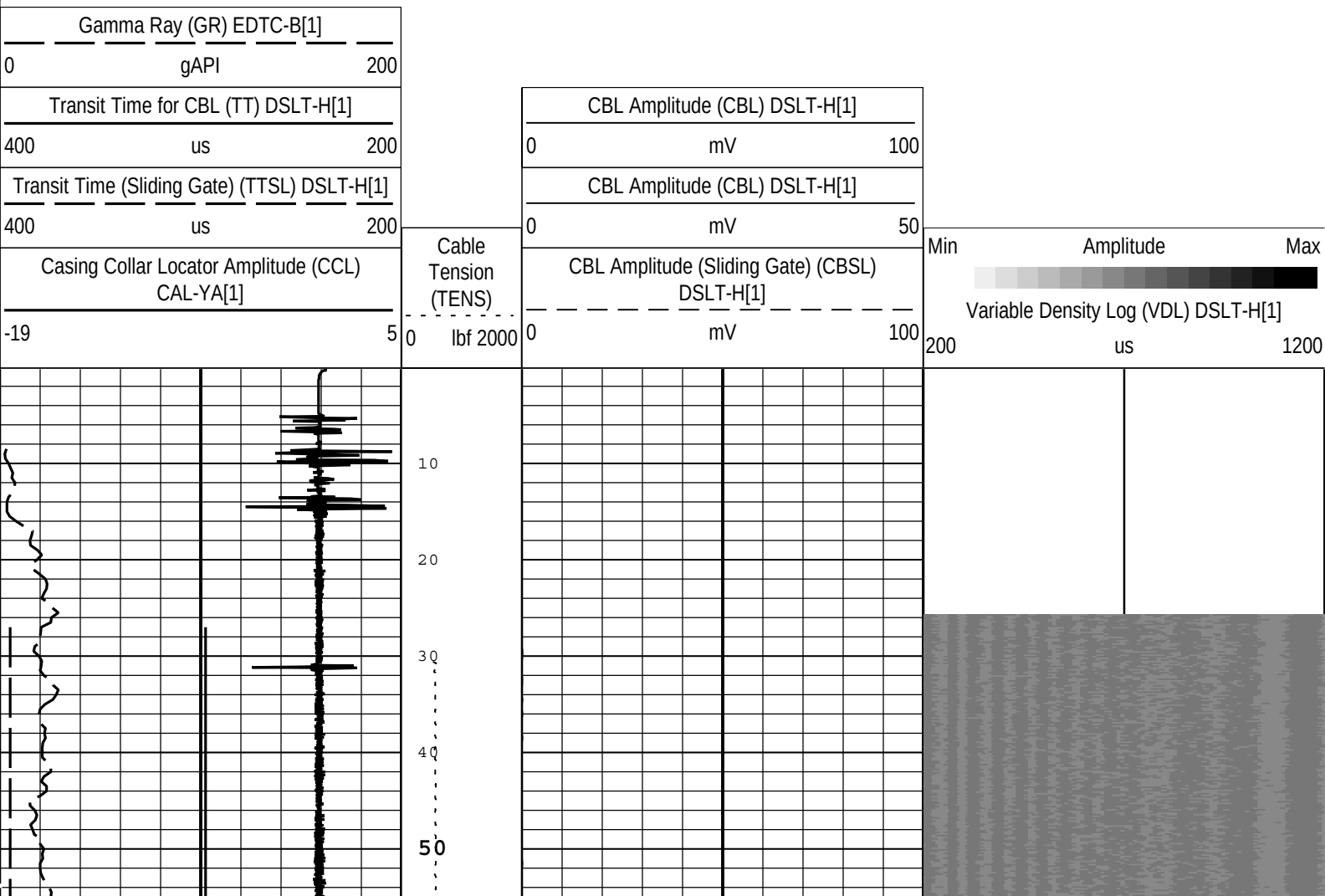
Company:BOB PT. BUMI SIAK PUSAKO - PERTAMINA HULU Well:PDD-M01 (# 156)

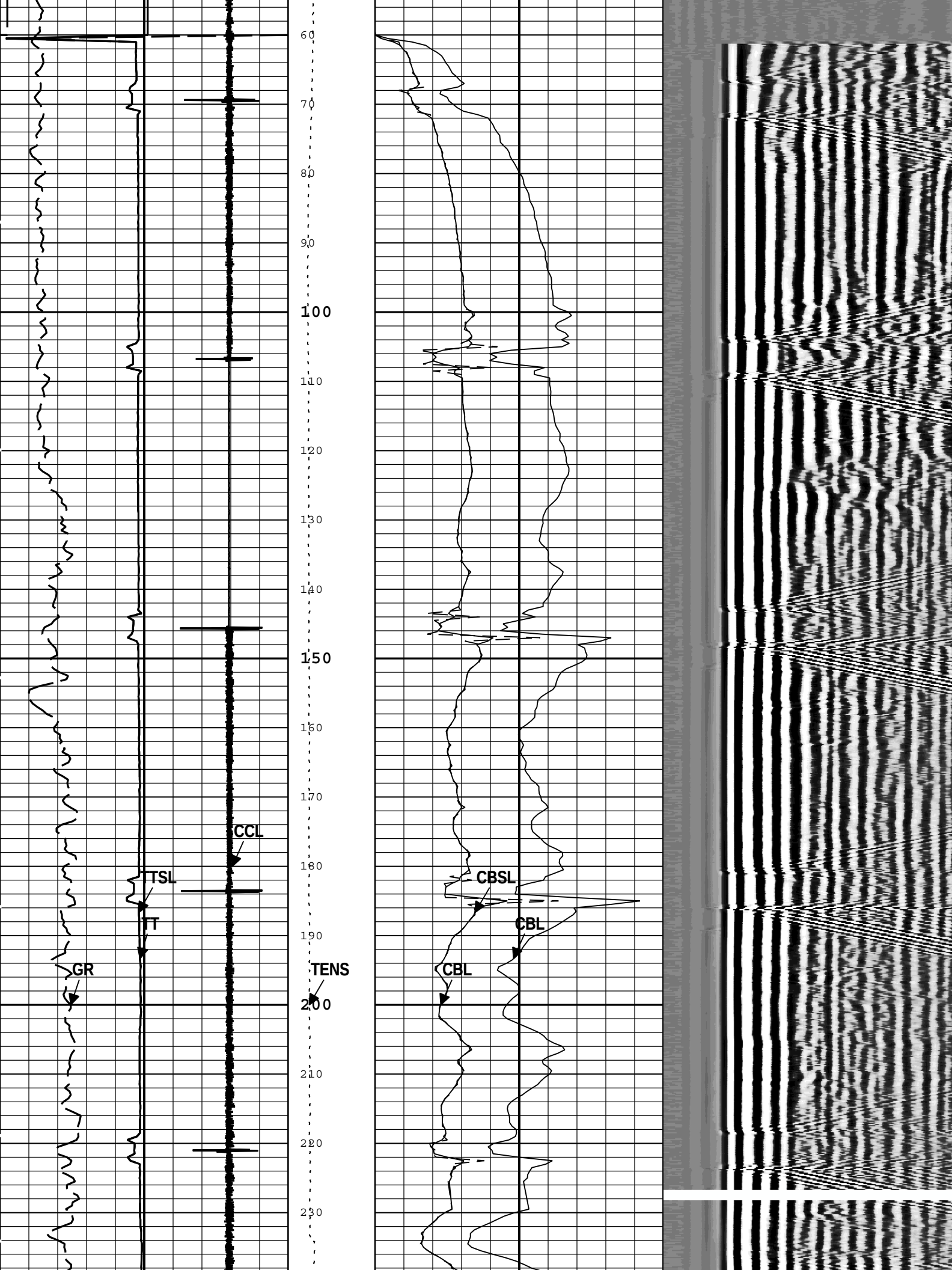
Composite 3:S035

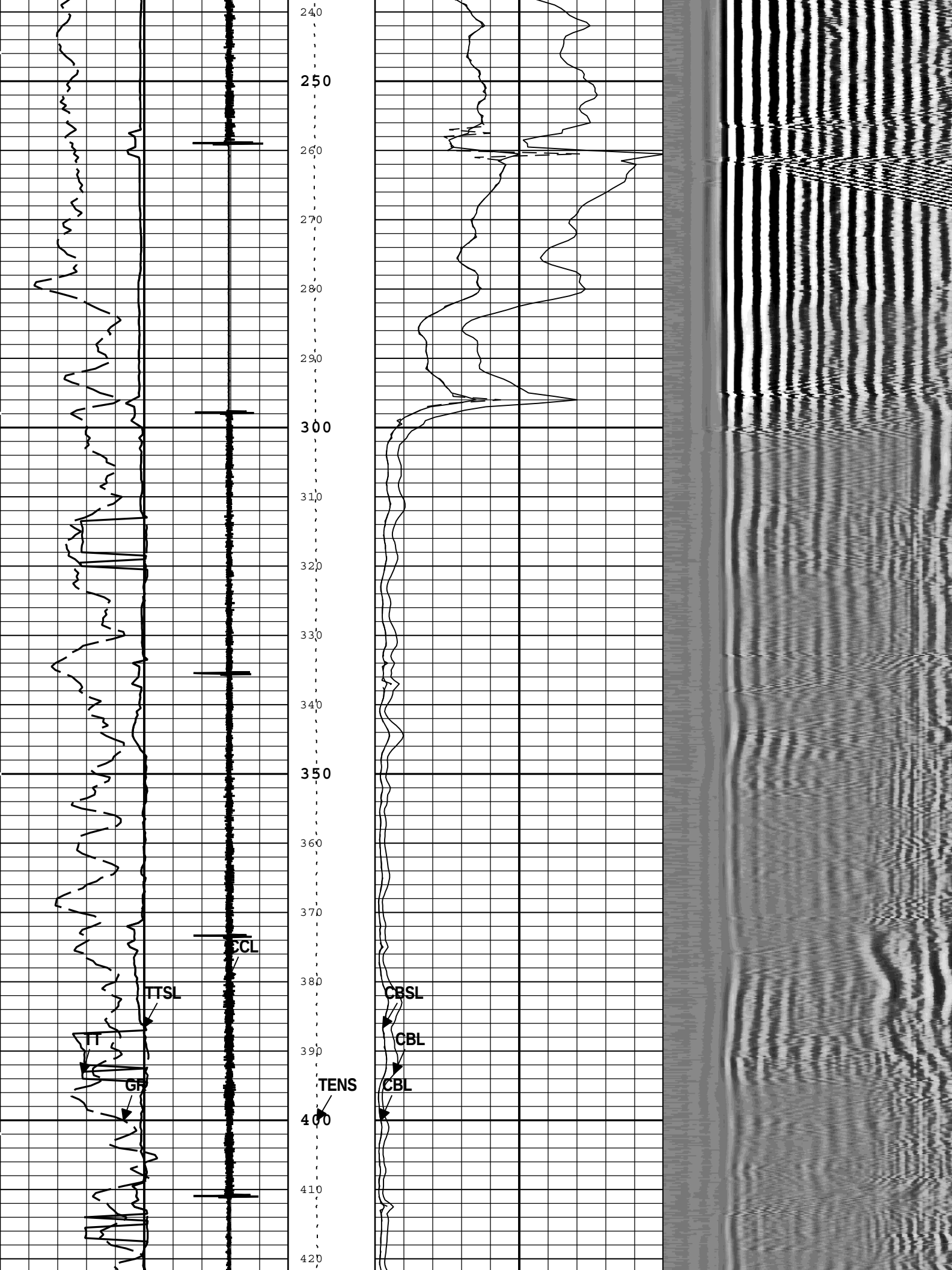
Description: CBL_VDL Format: Log (Sonic CBL with VDL) Index Scale: 6 in per 100 ft Index Unit: ft Index Type: Measured Depth Creation Date: 17-Oct-2014 20:20:41

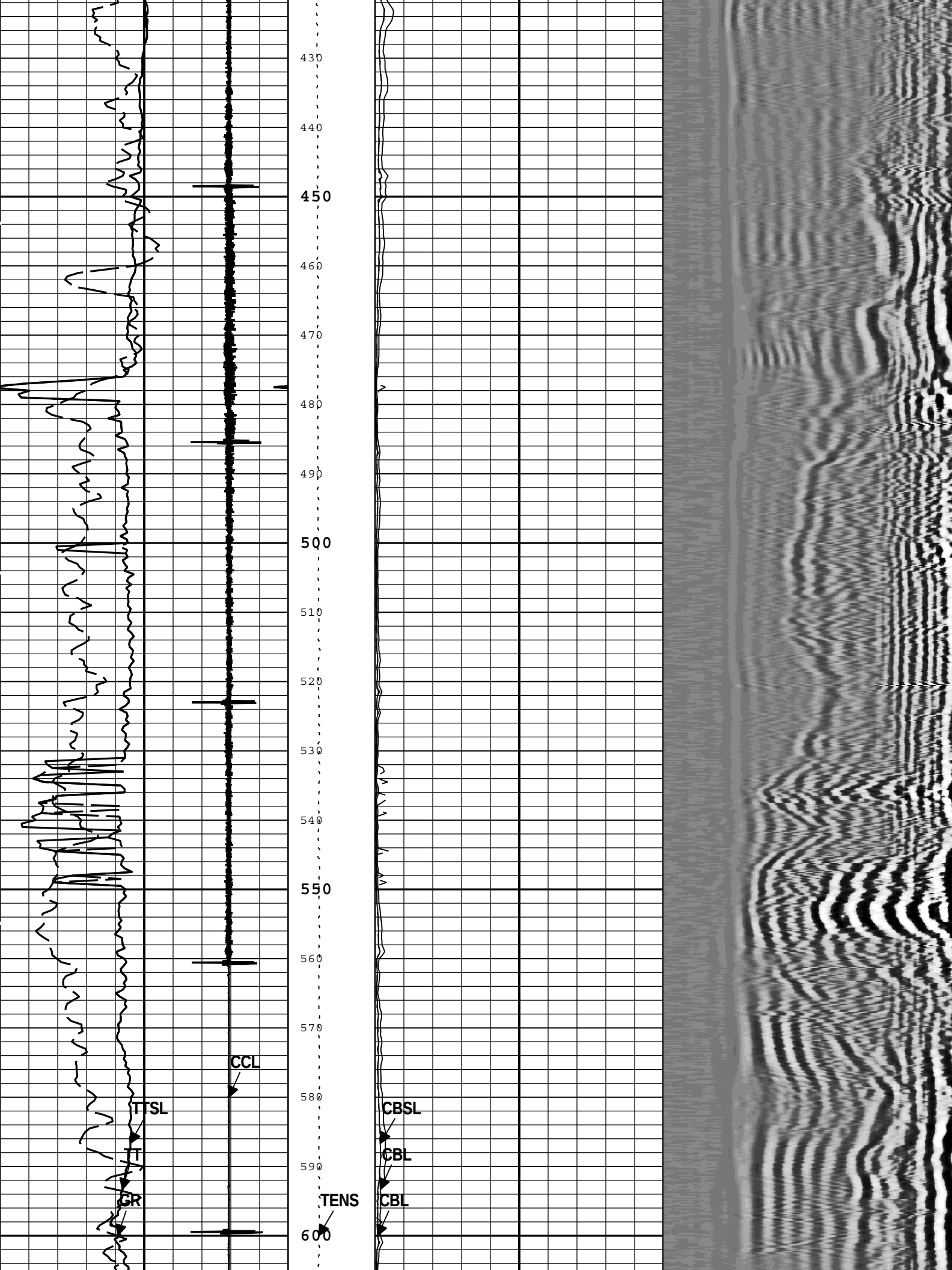
Channel	Source	Sampling
CBSL	DSLT-H[1]:SLS-E[1]:SLS-E[1]	6in
TIME_1900	WLWorkflow	0.1in
CBL	DSLT-H[1]:SLS-E[1]:SLS-E[1]	6in
CCL	CAL-YA[1]:CAL-YA[1]:CAL-YA[1]	1in
GR	EDTC-B[1]:EDTC-B[1]:EDTC-B[1]	6in
TENS	WLWorkflow	1in
TT	DSLT-H[1]:SLS-E[1]:SLS-E[1]	6in
TTSL	DSLT-H[1]:SLS-E[1]:SLS-E[1]	6in

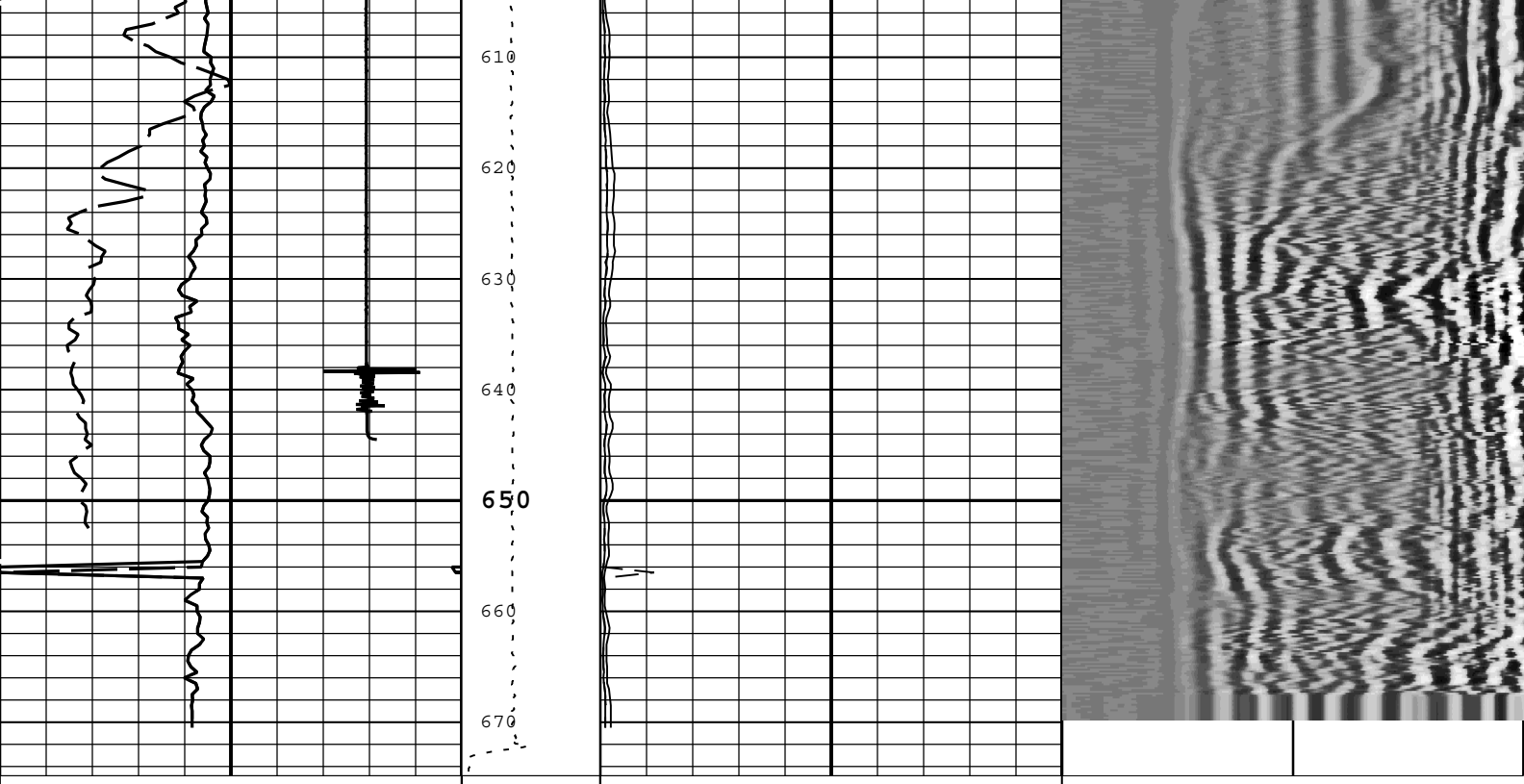
TIME_1900 - Time Marked every 60.00 (s)




60
70
80
90
100
110
120
130
140
150
160
170
180
190
200
210
220
230
CCL
TSL
TT
GR
TENS
CBSL
CBL
CBL


240
250
260
270
280
290
300
310
320
330
340
350
360
370
380
390
400
410
420
CCL
TTSL
TT
GR
TENS
CBSL
CBL
CBL


430
440
450
460
470
480
490
500
510
520
530
540
550
560
570
580
590
600
CCL
TTSL
TT
GR
TENS
CBSL
CBL
CBL



Gamma Ray (GR) EDTC-B[1]		
0	gAPI	200

Transit Time for CBL (TT) DSLT-H[1]		
400	us	200

Transit Time (Sliding Gate) (TTSL) DSLT-H[1]		
400	us	200

Casing Collar Locator Amplitude (CCL) CAL-YA[1]	
-19	5

Cable Tension (TENS)	
0	lbf 2000

CBL Amplitude (CBL) DSLT-H[1]		
0	mV	100

CBL Amplitude (CBL) DSLT-H[1]		
0	mV	50

CBL Amplitude (Sliding Gate) (CBSL) DSLT-H[1]		
0	mV	100

Min	Amplitude	Max

Variable Density Log (VDL) DSLT-H[1]

200	us	1200

TIME_1900 - Time Marked every 60.00 (s)

Description: CBL_VDL Format: Log (Sonic CBL with VDL) Index Scale: 6 in per 100 ft Index Unit: ft Index Type: Measured Depth Creation Date: 17-Oct-2014 20:20:41

1

Repeat Analysis 1:200

Pass Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	DSC Mode	Depth Shift	Include Parallel Data
1	Log[3]:Up	Up	232.97 ft	674.82 ft	17-Oct-2014 7:19:23 PM	17-Oct-2014 7:28:15 PM	ON	-0.60 ft	No
1	Log[4]:Up	Up	264.92 ft	576.51 ft	17-Oct-2014 7:30:17 PM	17-Oct-2014 7:36:23 PM	ON	-0.60 ft	No

All depths are referenced to toolstring zero

Log

Company:BOB PT. BUMI SIAK PUSAKO - PERTAMINA HULU Well:PDD-M01 (# 156)

1: Log[3]:Up:S035

Description: CBL_VDL Format: Log (Sonic CBL with VDL RA) Index Scale: 6 in per 100 ft Index Unit: ft Index Type: Measured Depth Creation Date: 17-Oct-2014 20:20:43

TIME_1900 - Time Marked every 60.00 (s)

Main To Repeat

Repeat To Main

Transit Time (Sliding Gate) (TTSL) DSLT-H

400 us 200

Main To Repeat

Repeat To Main

Casing Collar Locator Amplitude (CCL) CAL-YA

-19 5

Main To Repeat

Repeat To Main

Gamma Ray (GR) EDTC-B

0 gAPI 200

Main To Repeat

Repeat To Main

Transit Time for CBL (TT) DSLT-H

400 us 200

Main To Repeat

Repeat To Main

Cable Tension (TENS)

0 lbf 2000

Main To Repeat

Repeat To Main

CBL Amplitude (Sliding Gate) (CBSL) DSLT-H

0 mV 100

Main To Repeat

Repeat To Main

CBL Amplitude (CBL) DSLT-H

0 mV 100

Main To Repeat

Repeat To Main

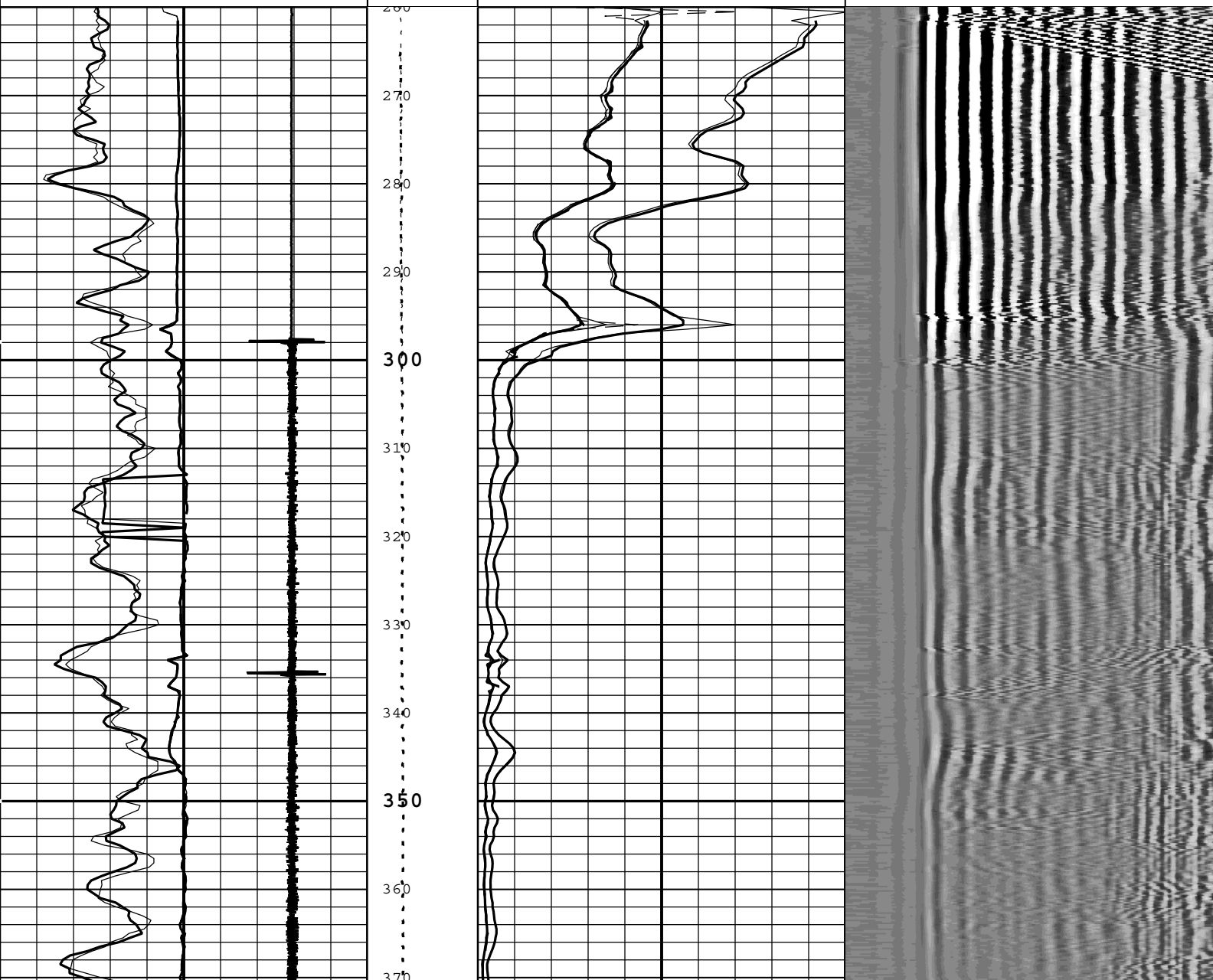
CBL Amplitude (CBL) DSLT-H

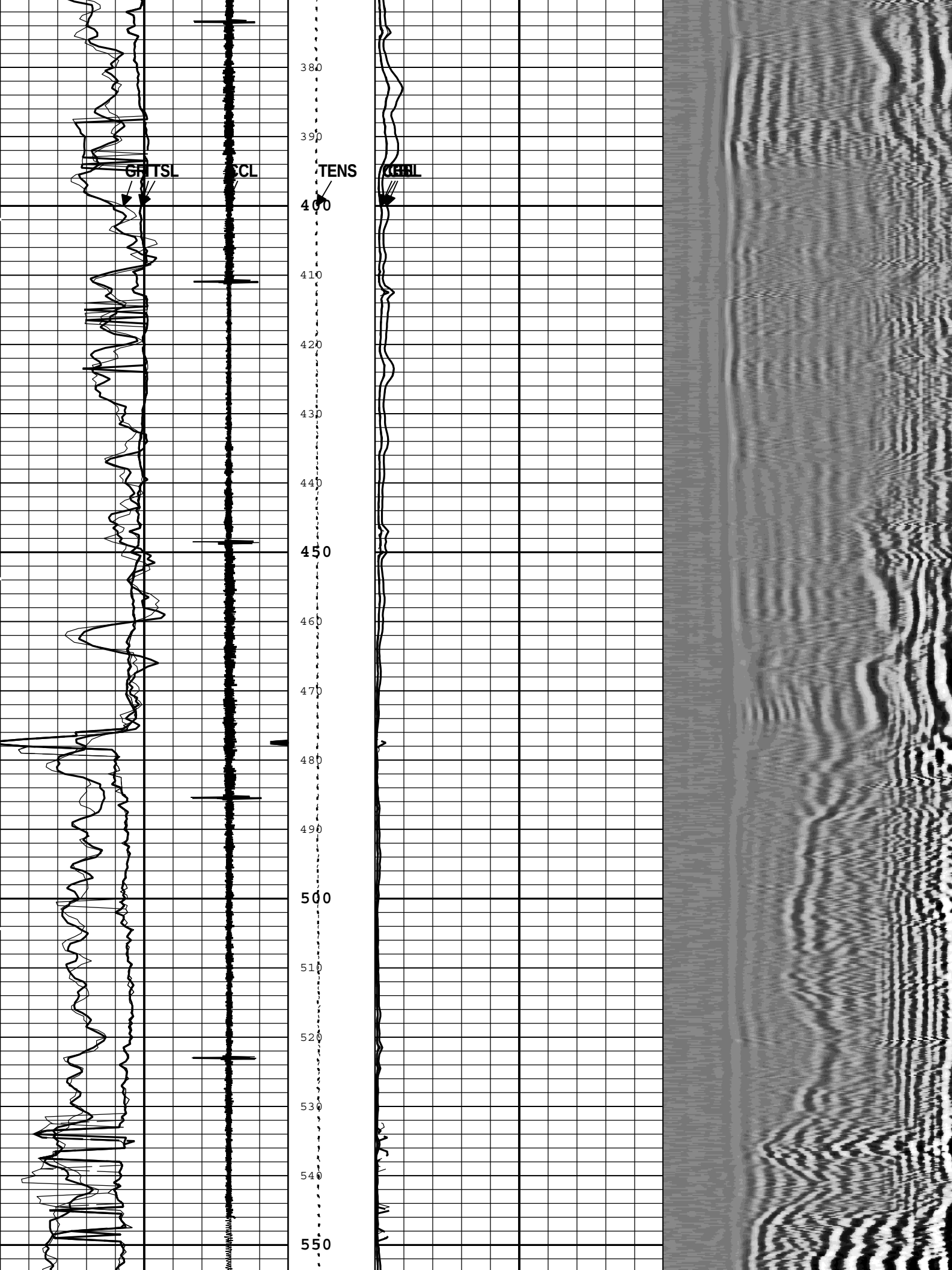
0 mV 50

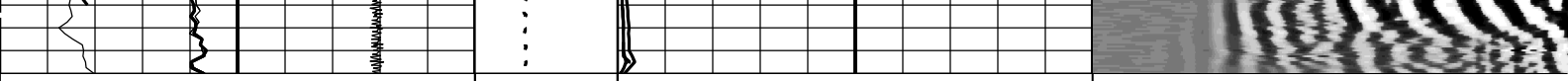
Min Amplitude Max

Variable Density Log (VDL) DSLT-H

200 us 1200




GRTTSL
CCL
TENS
CCBL
380
390
400
410
420
430
440
450
460
470
480
490
500
510
520
530
540
550



Main To Repeat

Repeat To Main

Transit Time (Sliding Gate) (TTSL) DSLT-H

400 us 200

Main To Repeat

Repeat To Main

Cable Tension (TENS)

0 lbf 2000

Main To Repeat

Repeat To Main

CBL Amplitude (Sliding Gate) (CBSL) DSLT-H

0 mV 100

Min Amplitude Max

Variable Density Log (VDL) DSLT-H

200 us 1200

Main To Repeat

Repeat To Main

Casing Collar Locator Amplitude (CCL) CAL-YA

-19 5

Main To Repeat

Repeat To Main

CBL Amplitude (CBL) DSLT-H

0 mV 100

Main To Repeat

Repeat To Main

Gamma Ray (GR) EDTC-B

0 gAPI 200

Main To Repeat

Repeat To Main

CBL Amplitude (CBL) DSLT-H

0 mV 50

Main To Repeat

Repeat To Main

Transit Time for CBL (TT) DSLT-H

400 us 200

TIME_1900 - Time Marked every 60.00 (s)

Description: CBL_VDL Format: Log (Sonic CBL with VDL RA) Index Scale: 6 in per 100 ft Index Unit: ft Index Type: Measured Depth Creation Date: 17-Oct-2014 20:20:43

Company:	BOB PT. BUMI SIAK PUSAKO - PERTAMINA HUL
Well:	PDD-M01 (# 156)
Field:	PEDADA
Rig Name:	APS-552
Country:	INDONESIA

Schlumberger

CBL-VDL-GR-CCL LOG

17-OCT-2014

SCALE 1:200